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Vegetation was not uniformly in equilibrium with climate during a period of climate stability

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Open research statement: Vegetation data are already published and publicly available (Dawson et al., 2016). Downscaled climate simulations are in preparation and available upon request. The code is provided in an external repository (https://github.com/amwillson/GJAM_STEPPS/).

Key words: climate sensitivity; ecotone; species range; Holocene; paleoecology; feedbacks

1 Abstract

Understanding how vegetation responds to climate is critical for anticipating the ecological impacts of anthropogenic climate change. While vegetation is often assumed to be in dynamic equilibrium with climate during periods of climate stability, evidence suggests that vegetation can also exhibit non-equilibrium responses to climatic shifts. Here, we investigate the prevalence of three classes of vegetation change—-equilibrium, lagged, and amplified-—in the Upper Midwest, USA, during the last 2,000 years of the pre-Industrial Holocene, a period of relatively stable climate. Using fossil pollen-derived vegetation reconstructions with fully specified uncertainty and new, downscaled climate simulations, we investigate the climate sensitivity of tree community composition. We find that vegetation predominantly exhibited equilibrium responses to climate. Further, we find no evidence of lagged vegetation change during this period of climatic stability. However, we identify evidence for amplified vegetation change at boundaries of the savanna biome and within hemlock-dominated forests at the western range limit of hemlock, wherein vegetation variability was driven by a combination of climate and vegetation-environment feedbacks. These findings highlight that even under relatively stable climate conditions, vegetation is not always in dynamic equilibrium with climate. Our results suggest that amplified vegetation change will be critical to consider under future, more rapid climate change, with implications for ecosystem management, conservation, and Earth system modeling.

2 Introduction

Understanding the sensitivity of vegetation to broad-scale changes in climate is critical for anticipating the effects of anthropogenic climate change on the distribution and abundance of plant species (Grierson et al., 2011; Armstrong et al., 2023). In practice, vegetation is often assumed to be in dynamic equilibrium with climate, wherein vegetation change is proportional to and concomitant with climate change over space and time (Williams et al., 2020; Whittaker, 1975; Toot et al., 2020) (Box 1; Figure 1a). However, vegetation can exhibit non-equilibrium responses to climate (Williams et al., 2020), including lagged and amplified responses (Figure 1a). Lagged vegetation change occurs when vegetation change is slower than, but proportional to, the pace of climate change, as a result of processes such as slow demographic rates in long-lived vegetation (e.g., trees) and dispersal limitations (Davis et al., 1986; Davis and Botkin, 1985; Holt, 2009; Williams et al., 2020; Talluto et al., 2017) (Box 1). Amplified vegetation change is characterized by vegetation change disproportional to, and potentially nonlinear with respect to, the pace of climate change (Williams et al., 2020, 2011; Ratajczak et al., 2018) (Box 1; Figure 1a). For instance, vegetation-mediated disturbances, where distinct vegetation communities promote different disturbance regimes (e.g., fire, herbivory), maintain distinct tropical savanna and forest biomes, and change to these disturbance regimes has plausibly led to fast, irreversible conversion of extant tropical forest to savanna in South America (Flores et al., 2024; Nobre et al., 2016). Similarly, the suite of vegetation-environment feedbacks occurring in closed-canopy forests, where the community alters the local environment (e.g., via microclimate buffering), make the environment more suitable for that forest community (Van Breugel et al., 2024; Milenkowic, 2024; Frelich and Reich, 1999).

Paleoecological records offer a source of information for understanding the climate-vegetation relationship over timescales sufficiently long for long-lived vegetation to respond to climate and for the consequences of vegetation-environment feedbacks to become apparent (Gajewski, 1993; Sanchez Goñi, 2022; Brown et al., 2025). Investigations of vegetation change in the early- to mid-Holocene have suggested that while many North American and European tree species tracked climate change during the rapid warming period of the early Holocene (i.e., dynamic equilibrium) (Williams et al., 2020; Davis and Shaw, 2001; Prentice et al., 1991), others lagged (Davis et al., 1998; Woods and Davis, 1989) or experienced amplified change with respect to climate forcing (Williams et al., 2009). Therefore, disequilibrium between climate and vegetation is generally accepted during periods of rapid climate change (Williams et al., 2020). It remains unclear under what circumstances lagged and amplified vegetation changes occur when climate change is more moderate.

The late Holocene offers a particularly relevant period during which to investigate the climate-vegetation relationship. Although discrete events and modest variability in climate occurred during the two millennia immediately preceding the industrial era (Calcote et al., 2021; Hotchkiss et al., 2007; Shuman and Marsicek, 2016), and land use also affected vegetation (Abrams and Nowacki, 2021; Briere and Gajewski, 2023; Munoz et al., 2014; Felde et al., 2024), both climate and vegetation were relatively stable over much of the world (Bonan, 2016; Elsig et al., 2009; PAGES 2k Consortium, 2013; Schimel et al., 2013; Williams et al., 2004). Vegetation is therefore often assumed to have been in dynamic equilibrium with climate during this pre-industrial era (Prentice et al., 1991; Schimel et al., 2013; Shuman et al., 2019). This assumption has implications for constructing models that anticipate the impact of climate change on vegetation. For instance, the pre-industrial climate-vegetation relationship is used (1) as a stable benchmark for species distribution (climate envelope) models (Bickford and Mackey, 2004; Iverson et al., 2019; Jackson and Williams, 2004; Schussman et al., 2006), (2) to measure the impact of industrial-era climate change in Earth system models spun-up to equilibrium (Dallmeyer et al., 2021; Albani et al., 2006), and (3) to establish management targets for restoration consistent with species' historical climate niches (Grace et al., 2019; Williams et al., 2020; Barnosky et al., 2017; Booth et al., 2023; Swetnam et al., 1999). However, quantitative assessments of the late Holocene relationship between climate and vegetation are limited because of a lack of (1) quantitative reconstructions with uncertainty of tree distributions and abundances (rather than fossil pollen proportions) and (2) independent reconstructions of climate over paleoecological timescales (Marlon et al., 2017; Paciorek and McLachlan, 2009).

The Upper Midwest, USA (UMW), a region with rich paleoecological records, offers evidence that the dynamic equilibrium assumption may be inappropriate, even during the late Holocene (Figure 1b). Despite the fact that both climate and vegetation were relatively stable during the late Holocene, recent reconstructions of vegetation community composition in the UMW during this period identified notable changes in community composition both at the savanna-forest boundary and within the northern mixedwood forests of northern Wisconsin (Dawson et al., 2016, 2019) (Figure 1b). The savanna-forest boundary, where oak (*Quercus*)-dominated savannas met closed forest of pine (*Pinus*) or northern hardwoods, shifted westward (McAndrews, 1966; Jacobson, 1979; Nelson and Hu, 2008), while the northern mixedwood forests experienced an increase in the relative abundance of eastern hemlock (*Tsuga canadensis*) near its western range margin (Davis et al., 1986). Against the regional background of negligible change in community composition, the observations of discrete compositional change along ecotone and species boundaries in Dawson et al. (2019) raise questions about the role of climate in driving vegetation change. Both changes are consistent with paleoecological trends toward cooler and moister conditions (Marlon et al., 2017), but Dawson et al. (2019) did not formally assess the sensitivity of vegetation to climate because, until recently, independent reconstructions of climate and vegetation were unavailable (Marlon et al., 2017; Paciorek and McLachlan, 2009).

Here, we examine the climate sensitivity of vegetation by combining the community composition reconstructions, with uncertainty, from Dawson et al. (2016) with new, independent reconstructions of climate conditions. We use a series of statistical models to evaluate vegetation trends in the UMW for changes consistent with equilibrium, lagged, and amplified responses to climate. Given the relative stability of climate and vegetation during the late Holocene, we hypothesize that vegetation was primarily characterized by equilibrium responses. Furthermore, because climate had been stable for centuries, we hypothesize that unusually large changes in community composition at the savanna-forest ecotone and at the range limit of hemlock is not consistent with lagged vegetation responses to past climate change. Instead, we hypothesize that these changes in vegetation are more compatible with the amplified response type, in line with previous work highlighting the role of internal feedbacks in the UMW during the early Holocene (Williams et al., 2009).

3 Methods

3.1 Data and simulations

3.1.1 Community composition

We used fossil pollen-derived reconstructions of tree community composition from the Spatio-Temporal Empirical Prediction from Pollen in Sediments (STEPPS) pollen-vegetation model developed by Dawson et al. (2016) and Dawson et al. (2019) for the UMW. The composition data were relative abundance estimates of nine dominant and ecologically important tree taxa on a 24 km x 24 km grid across Minnesota, Wisconsin, and Upper Michigan at 100 year intervals from 1,900 to 300 years before 1950 (YBP). The taxa that were included in our analysis are beech (*Fagus grandifolia*), birch (*Betula spp.*), elm (*Ulmus spp.*), hemlock, maple (*Acer spp.*), oak (*Quercus spp.*), pine (*Pinus spp.*), spruce (*Picea spp.*), and tamarack (*Larix laricina*). These reconstructions are consistent with previous vegetation histories of the region (Goring et al., 2016; Curtis, 1959): southern hardwood forests to the east consist primarily of maples and pines (on sandy soils); northern mixedwood forests to the north, of conifers (pines, hemlock, and tamarack) and birches; and open-canopy savannas to the southwest, of oaks (Figure 1b). Because our reconstructions are only of tree taxa, the southwest corner of Minnesota shows a signal of high elm relative abundance because elm occurred along river corridors in an area otherwise dominated by tallgrass prairie (Goring et al., 2016; Grimm, 1984; Williams et al., 2009) (Figure 1b). The relative abundances of two additional groups, "other conifer" (most relatively abundant in northern mixedwood forests) and "other hardwood" (most relatively abundant along the savanna ecotones) were also estimated and encompass all rare taxa in either broad taxonomic grouping. Maps of our community composition reconstructions can be found in our online supplementary materials (`www.github.com/amwillson/GJAM_STEPPS/full_supplement.pdf`).

3.1.2 Climate and soil

We derived independent statistically downscaled reconstructions of climate conditions (Guaita et al., *in prep.*) using the "past2k" experiment from the Paleoclimate Model Intercomparison Project (PMIP4) (Jungclaus et al., 2017). At the time of analysis, only the Max Planck Institute Earth System Model (MPI-ESM1.2) (Mauritsen et al., 2019) had published climate reconstructions under the past2k experiment (Jungclaus et al., 2021). The simulations are modeled, not data-derived. MPI-ESM1.2 has been validated, but not in detail and not for our region; therefore, these simulations miss discrete events and trends (i.e., the Holocene Conundrum) (Bader et al., 2020; Kaufman and Broadman, 2023). The trends and magnitude of change in the reconstructions we use are broadly similar to those from empirical paleoclimate data (Dallmeyer et al., 2021; Jungclaus et al., 2021), but discrepancies between simulated and proxy-derived climate estimates suggest that model simulations may underestimate directional trends in climate during the late Holocene (Kaufman and Broadman, 2023). Nevertheless, empirical reconstructions that are independent from fossil pollen records do not have the coverage or resolution to fill the role that modeled climate fills in this study.

We used Principal Component Regression (PCR) (Jolliffe, 2002) to downscale the climate reconstructions from the 1.875 x 1.875 degree resolution of PMIP4 to a 0.08 x 0.08 degree resolution, aligning with the spatial resolution of our vegetation reconstructions (Guaita *in review*). The regression was fit between synoptic simulations from the "historical" experiment (1850-2015) from MPI-ESM1.2 (Eyring et al., 2016) and local-scale climate data from NOAA and CANADA measurement networks (Lawrimore et al., 2016; Environment and Climate Change Canada, 2022). The fitted relationship was then applied back in time to downscale climate simulations from the past2k experiment. Using the downscaled climate simulations, we estimated average annual temperature (annual mean of monthly temperature), total annual precipitation (annual sum of monthly precipitation), intraannual temperature seasonality (annual standard deviation of monthly temperature), and intraannual precipitation seasonality (annual standard deviation of monthly precipitation), which previous work has found to be the principal drivers of vegetation over paleoecological time scales (Davis and Shaw, 2001; Hupy, 2012; Curtis, 1959; Umbanhowar et al., 2006; Nelson and Hu, 2008; Abrams, 1990; Novick et al., 2022). Our reconstructions show limited directional change in climate over time in our study domain.

In addition to climate, we included soil texture in our model because soil texture modifies the impact of temperature and precipitation on vegetation (Saxton et al., 1986). We used the gridded Soil Survey Geographic Database (gSSURGO) produced by the U.S. Department of Agriculture Natural Resources Conservation Service (Soil Survey Staff, 2023c,b,a). We downloaded the gSSURGO data products and extracted the map unit properties of soil % sand and soil % silt using the "dominant component (numeric)" method and 0-30 cm soil depth at 700 m resolution using the soilDB R package (Beaudette et al., 2024), then aggregated the soil texture estimates to our 24 x 24 km grid. The gSSURGO database includes data from the modern observational period, but we expect that change in soil texture has been negligible over 2,000 years because of the slow rate of soil development. Maps of all our environmental covariates can be found in our online supplementary materials (`www.github.com/amwillson/GJAM_STEPPS/full_supplement.pdf`).

3.2 Analytical approach

We conducted a series of complementary analyses to identify equilibrium, lagged, and amplified vegetation changes. We used our analyses to draw conclusions about the likely relationships between climate and vegetation at coarse resolutions, recognizing that the high uncertainty in paleoecological reconstructions and the possibility of confounding variables precludes definitive judgments of climate-vegetation relationships. To assess the extent to which vegetation followed the equilibrium response type, we fit the relationship between climate and community composition using a Joint Species Distribution Model (JSDM; Section 3.2.1). Correlative SDMs implicitly assume that vegetation is in equilibrium with climate. The model's residuals represent variation in community composition that was not explained by the climate variables included in our model, and correlations between taxon residuals quantify patterns in community composition not explained by climate (Clark et al., 2017). Smaller residual correlations are indicative of equilibrium vegetation dynamics, while larger residual correlations are indicative of lagged or amplified (i.e., non-equilibrium) vegetation responses (Figure 1c).

Next, we examined instances where our JSDM fails to explain the observed climate-vegetation relationship. To do so, we first assessed the alternative that vegetation is responding more slowly to climate than the centennial resolution of our model. To evaluate this, we tested whether including previous climate conditions improved predictions of community composition (Section 3.2.2). Improved prediction accuracy from including previous climate conditions would be indicative of the lagged vegetation response type (Figure 1d). We then investigated the rate of vegetation change relative to the rate of climate change over time for each taxon and each grid cell in our domain ("climate sensitivity" of vegetation; Section 3.2.3). Relative to the background rate of low sensitivity resulting from limited climate change, higher climate sensitivity indicates that community composition changed more than anticipated by changing climate alone (Williams et al., 2011, 2020). This points toward amplified vegetation change, where change in community composition is driven by vegetation-environment feedbacks in addition to external climate forcing (Figure 1e). Finally, we built an autoregressive model to investigate the phenomenon of rising hemlock relative abundance in northern Wisconsin, which was not explained by the previous analyses (Section 3.2.4). Higher sensitivity to previous community composition compared to climate conditions would support the amplified vegetation response because this would indicate that hemlock relative abundance was more related to its previous relative abundance than changes in climate (Figure 1f).

3.2.1 Joint Species Distribution Model

We used the Generalized Joint Attribute Model (GJAM) as our JSDM using the GJAM R package (Clark et al., 2017). GJAM is a hierarchical Bayesian JSDM, where the climate-vegetation relationships are modeled as linear with respect to the parameters. We began by fitting GJAM to the relationship between climate (and soil) conditions and tree community composition at the same time step with no interaction terms. This is comparable to JSDM applications assuming an equilibrium vegetation response to climate change (Sandel et al., 2025). We used taxon relative abundances as the response variables, specifying the "fractional composition" response data type.

The model estimates three classes of parameters. First is the relationship between each environmental (climate and soil) driver and each response variable. Second is the joint sensitivity of all response variables to each independent variable ($\hat{F}$), which is a function of the environment-vegetation coefficients and the covariance matrix. Higher sensitivity values mean that the independent variable has a stronger relationship with community composition overall than independent variables with lower sensitivity. Third is the residual covariance, which quantifies patterns in the response variables after accounting for their joint dependence on the independent variables, and was used to calculate Pearson correlation matrices. We interpret neutral correlations as indicating that the independent variables in our model were sufficient to explain patterns in community composition, while non-neutral correlations indicate that the included independent variables were insufficient to explain patterns in community composition (Clark et al., 2017). Non-neutral correlations can arise because of either missing climate covariates that are important for explaining community composition, or missing processes such as vegetation-environment feedbacks. For instance, positive correlations may indicate that in the same climate and soil conditions, the taxa are more likely to be abundant together (from, e.g., facilitation) and negative correlations may indicate that when one taxon is abundant, the other is likely to be less abundant even in the same climate conditions (from, e.g., feedbacks leading to distinct communities under similar climate conditions). In this way, neutral residual correlations provide evidence supporting the equilibrium vegetation response type, while large, non-neutral correlations support the existence of non-equilibrium (lagged or amplified) vegetation responses (Figure 1c) when combined with other lines of evidence (Figure 1d-f).

Model fitting To avoid incorrect attribution of ecological meaning to the spatial and temporal autocorrelation induced by the spatiotemporal smoothing in the STEPPS model, we subsampled the grid cells used for fitting the model in space and time (Supplementary Methods 1.1). Our subsampling method left us with 80 evenly-spaced 24-km grid cells in each of five evenly-spaced time steps (1900, 1500, 1100, 700, and 300 YBP; n = 400 grid cells). The remaining grid cells (n = 11,568) were used for model validation (Section 3.2.1). Among our independent variables, our two metrics of soil texture (soil % sand and soil % silt) and two of our climate variables (temperature seasonality and total annual precipitation) were correlated. Therefore, we performed model selection to identify the model that best captures spatiotemporal trends in community composition with relatively uncorrelated covariates (Supplementary Methods 1.2; Supplementary Table S1). Following model selection, we chose a model with the following covariates: average annual temperature, total annual precipitation, precipitation seasonality, and soil % sand with no interaction terms, although our results changed negligibly regardless of which covariates were used (Supplementary Figures S1-S6) or if interactions were included (Supplementary Figures S20-S21).

We fit GJAM 100 times to each of 100 posterior draws from the STEPPS model and pooled the parameter estimates across all 100 model fits. This allowed us to account for the uncertainty in the vegetation reconstructions in our parameter estimates. For each model, we ran the Markov Chain Monte Carlo (MCMC) algorithm for 10,000 iterations, removing the first 2,000 iterations for burn-in. We used one Markov chain, which is the only option in the GJAM package (Clark et al., 2017). We used trace plots for each parameter to assess the model for convergence, with all parameters converging after the burn-in period.

Model validation We performed model validation in two phases. First, we fit our model to the first four time steps (1900, 1500, 1100, and 700 YBP; n = 320 grid cells) and withheld the same grid cells in the final time step (300 YBP; n = 80 grid cells) as out-of-sample validation ("temporal validation"). We then predicted community composition given the independent variables and the latent covariance between taxa using the gjamPredict function from the gjam package (Clark et al., 2017). We compared estimated ("observed") and predicted relative abundances of each taxon at each grid cell in the 300 YBP time step. The purpose of our temporal validation was to ascertain the degree to which our model was overfit to our subsampled data by predicting a different time period (Supplementary Methods 1.3). Once the temporal validation was completed, we refit our final model including the relative abundances from 300 YBP. Then, we used the fitted model to predict all the spatiotemporal estimates not used to fit the model. This final out-of-sample dataset included 11,568 observations across 17 time steps and 704 spatial locations ("spatiotemporal validation"). We assessed prediction accuracy of our final model by comparing predictions to our empirical reconstructions across the entire spatiotemporal domain of the community composition reconstructions (Supplementary Methods 1.3).

3.2.2 Effect of lagged climate variables in GJAM

To test whether tree community composition exhibited lagged response to climate, we included climate variables from the previous (i.e., 100 years prior) and current time steps as independent variables in our GJAM ("lagged climate model"). As with our previous GJAM, we fit the model 100 times using different posterior draws of our community composition reconstructions, ran the MCMC algorithm for 10,000 iterations, removing the first 2,000 iterations for burn-in, and visually assessed parameters for convergence. All parameters appeared to converge, despite suspected strong trade-offs between autocorrelated climate variables. We then used the gjamPredict function to predict tree community composition in all out-of-sample grid cells.

We assessed whether lagged vegetation responses occurred by comparing the accuracy of out-of-sample predictions from our models with and without lagged climate covariates. Specifically, we compared the correlation between observed and predicted relative abundance of each taxon between models. If the correlations were higher with predictions from the lagged climate model, then we would infer that vegetation exhibited lagged response to climate change because including previous climate improved predictions (Figure 1d). If the correlations were approximately the same, that would indicate that previous climate did not improve the model's predictive ability, and vegetation did not experience lagged response to climate. If correlations were worse with predictions from the lagged climate model, this might indicate that the inclusion of previous climate variables led to overfitting the climate-vegetation relationship.

3.2.3 Vegetation sensitivity to climate change

We evaluated the climate sensitivity of vegetation to temporal changes in climate in each 24-km grid cell individually (Figure 1e). For each grid cell and each taxon, we fit a multiple linear regression model to the relationship between relative abundance and standardized average annual temperature, total annual precipitation, and precipitation seasonality. We pooled the posterior samples from the STEPPS data product such that each linear model was fit to 1700 observations (17 time steps x 100 posterior samples), allowing the uncertainty in the slopes to include the uncertainty in our vegetation reconstructions. We compared slope estimates over space and between taxa to investigate under what circumstances changes in climate led to greater changes in community composition. Slopes near zero are indicative of low temporal sensitivity, wherein a change in the climate variable did not lead to a change in the taxon's relative abundance. Meanwhile, non-zero slopes indicate that as climate changed over time, the taxon's relative abundance increased (positive slope) or decreased (negative slope). Because directional climate change and change in community composition were limited during our study period, we expect slopes to be near zero for most taxa across most of the region. Consistent, near-zero slopes across grid cells indicate that vegetation was in equilibrium with climate. Alternatively, higher absolute slopes in some grid cells relative to a background of slopes near zero indicates non-equilibrium, amplified vegetation response to climate (Figure 1e). The slopes in this analysis differ from the slopes in our GJAM because these slopes only consider the relationship between climate and community composition over time, rather than over space and time simultaneously.

3.2.4 Autoregressive model

Despite negligible change in climate in the UMW over the last two millennia, hemlock increased in relative abundance in north-central Wisconsin, and this increase was not predicted by our GJAM, leading to worsening prediction accuracy over time in this region. This rise in hemlock relative abundance was also not explained by our lagged climate model, nor did hemlock in this region show high climate sensitivity. One reasonable explanation for this phenomenon is that hemlock experienced amplified vegetation change independent of climate. We assessed for the possibility that hemlock experienced amplified change by building a Bayesian linear model with a first-order autoregressive term for only the hemlock taxon. The linear model used hemlock relative abundance as the response variable and five predictor variables: average annual temperature, total annual precipitation, precipitation seasonality, soil % sand, and the first-order autoregressive term. We expressed the independent variables in units of standard deviation to allow for direct comparisons between model coefficients. We fit this model two times: (1) using the 25 grid cells closest to the peak of hemlock relative abundance in north-central Wisconsin and (2) using the entire geographic domain of our analysis. If the change in hemlock relative abundance was driven by self-reinforcing feedbacks, then we would expect (1) previous hemlock abundance to be a better predictor of current hemlock abundance than climate or soil texture, and (2) the relationship between previous and current hemlock relative abundance to be stronger in the small domain in north-central Wisconsin where hemlock increased than across the entire UMW (Figure 1f). A stronger relationship between previous and current hemlock relative abundance in the smaller region in Wisconsin would suggest that subtle climate trends, which were overpowered by the spatial climate signal across our full domain, could not explain the rise in hemlock abundance in this region.

4 Results

4.1 Joint Species Distribution Model

4.1.1 Model validation

In general, our model was able to predict spatiotemporal patterns in community composition as a function of the contemporaneous climate and the latent covariance between taxa, according to our temporal validation (Supplementary Results 2.2; Supplementary Figure S7). When accounting for the uncertainty in community composition, prediction bias overlapped zero for most taxa in most grid cells (Supplementary Results 2.2). However, the model tended to strongly underpredict the relative abundance of all taxa where the taxon was most relatively abundant. Reciprocally, other taxa were often slightly overpredicted to compensate for the underprediction. Proportional data like our community composition are difficult to model with low bias because of the constraint that the data must sum to one, and the tendency of linear models like GJAM to make predictions close to the mean of the response variable and not capture more extreme values (Clark et al., 2017; Prentice et al., 1991). We assessed the likelihood that these limitations of the model contribute to underprediction in our validation by conducting another prediction experiment. We predicted the relative abundance of one taxon at a time, given both the climate and soil texture covariates and the relative abundances of all ten other taxa at the same location (Supplementary Methods 1.4). Because there is a sum to one constraint, the model should make accurate predictions even where taxa are highly relatively abundant. Instead, we found that relative abundances were still underpredicted when the taxon was highly abundant, even when all other relative abundances were known (Supplementary Results 2.4; Supplementary Figure S22). Therefore, we infer that our model is more suitable for inferring relationships between climate and vegetation, and correlations between taxa, rather than predicting the precise community composition. For full details and results of this experiment, we refer the reader to our online supplementary materials (`https://github.com/amwillson/GJAM_STEPPS/full_supplement.pdf`).

In our spatiotemporal validation, we found that our model captured general spatiotemporal trends in community composition (Supplementary Results 2.3). The median Pearson product-moment correlation (r) between empirically estimated and predicted relative abundance across all taxa was 0.68, with 9 of 11 taxa having r > 0.6. Maple was the worst-predicted taxon (r = 0.29) and oak was the best-predicted taxon (r = 0.76). Maple had a low correlation coefficient because the model predicted it to be at medium relative abundance across most of our study region, when in fact it was more relatively abundant in forested regions than in open-canopy regions. Additionally, our estimates of maple relative abundance are among the most uncertain in the STEPPS model owing to maple's entomophilous pollination strategy (Dawson et al., 2019), a known challenge in pollen-based paleoecological reconstructions (Jackson, 1994). Prediction bias tended to be near zero (i.e., high agreement between predicted and observed relative abundance) for all taxa in most grid cells (Supplementary Figure S8). The median prediction bias across space was generally low (-0.01 fraction of stems [spruce] to -0.07 fraction of stems [oak]), but showed the tendency of our GJAM to underpredict relative abundance where each taxon was most relatively abundant (Supplementary Figures S9-S19). The 95% credible intervals of prediction bias overlapped zero most of the time, suggesting that model inaccuracy was often a product of noise in the relative abundance estimates. We attributed both patterns of underprediction of most taxa at their peak relative abundances and overprediction of oak in regions adjacent to its peak relative abundance to the difficulty of predicting proportional data, based on our conclusions from the temporal validation. In these cases, the spatiotemporal patterns of relative abundance remained congruent with observed trends, but the magnitude of relative abundance was incorrect, indicating that our model is sufficient for examining spatiotemporal trends in community composition.

The only instance in which the model had high prediction error with worsening accuracy over time was hemlock in north-central Wisconsin, the same region identified as having anomalously large community composition change in Dawson et al. (2019). Hemlock increased in relative abundance in this region (Supplementary Figure S12), from approximately 25% of total stems to 62% of total stems between 1900 and 400 YBP. The model did not predict increasing relative abundance, leading to larger prediction bias over time (0.20 [1900 YBP] to 0.55 [300 YBP]). This indicates that the model was unable to explain both the trend and the magnitude of the increase in hemlock relative abundance in this region.

4.1.2 Contemporaneous climate explained overall patterns in community composition

As expected, contemporaneous climate and soil conditions explained large-scale patterns in community composition (Figure 2). Taxa were jointly most sensitive to average annual temperature (median $\hat{F}$ [95% credible interval (CrI)] = 100 [58,156]) and precipitation seasonality (median $\hat{F}$ [95% CrI] = 75 [47, 112]); Figure 2d). To a lesser extent, community composition was sensitive to total annual precipitation (median $\hat{F}$ [95% CrI] = 33 [22,64]) and soil texture (median $\hat{F}$ [95% CrI] = 12 [6,31]; Figure 2d). Average annual temperature explained changes in community composition occurring along the north-south gradient (Figure 2a). Birch, spruce, other conifer taxa, and tamarack had strong negative coefficients (median β [95% CrI] = -0.80 [-1.0,-0.59], -0.77 [-1.1,-0.48], -0.77 [-1.1,-0.44], and -0.75 [-1.0,-0.47], respectively), all of which were prevalent in northern mixed forests (Figure 1b). Meanwhile, oak and elm, the taxa with highest relative abundance in the southern portion of the UMW (Figure 1b), have strong positive coefficients (median β [95% CrI] = 0.94 [0.73,1.2] and 0.78 [0.56, 1.0], respectively). Similarly, total annual precipitation explained changes in community composition occurring along an east-west gradient (Figure 2b). Other hardwood taxa and elm, the taxa most abundant in the open canopy systems in the northwest and southwest portions of the UMW, respectively, were negatively associated with total annual precipitation (median β [95% CrI] = -0.77 [-1.1,-0.48] and -0.72 [-0.97,-0.49], respectively). Several forest taxa with high relative abundance in the east (hemlock, birch, and pine) were positively associated with total annual precipitation (median β [95% CrI] = 0.65 [0.48,0.82], 0.45 [0.26,0.65], and 0.42 [0.26,0.58], respectively). Precipitation seasonality was the covariate that most strongly differentiated forest taxa from savanna taxa (Figure 2c). Forest taxa, including beech, hemlock, birch, and other conifer taxa, were negatively related to precipitation seasonality (median β [95% CrI] = -0.57 [-0.76,-0.39], -0.37 [-0.54,-0.20], -0.37 [-0.56,-0.18], and -0.35 [-0.57, -0.16], respectively). Oak and, to a lesser extent, other hardwood taxa, were positively related to precipitation seasonality (median β [95% CrI] = 0.32 [0.18, 0.46] and 0.19 [0.017, 0.36], respectively), indicating that taxa most prevalent in the savanna were less susceptible to seasonal droughts.

4.1.3 Strong residual correlations existed between taxa

After accounting for their joint dependence on climate and soil texture, strong residual correlations remained between taxon relative abundances in two instances (Figure 3). First, the strongest residual correlations were between taxa associated with the savanna-forest (oak and maple median r [95% CrI] = -0.46 [-1.0,-0.19]) and savanna-prairie (oak and elm median r [95% CrI] = -0.56 [-1.0,-0.15]) ecotones, where maple, oak, and elm were the most relatively abundant tree taxa of the hardwood forest, savanna, and prairie biomes, respectively. Weaker negative residual correlations existed between oak and nearly every other tree taxon (Figure 3). Second, moderately strong residual correlations existed between hemlock and pine (median r [95% CrI] = -0.16 [-0.59,-0.020]), hemlock and spruce (median r [95% CrI] = -0.22 [-1.0,-0.034]), and hemlock and tamarack (median r [95% CrI] = -0.18 [-0.74,-0.029]), where pine, spruce, and tamarack were relatively abundant conifers in northern UMW forests. In contrast, most other taxon pairs exhibited minimal residual correlations, highlighting that climate conditions sufficiently explained patterns in community composition, except at ecotones and in Wisconsin hemlock-dominated forests. When different sets of climate and soil texture variables were used, similar patterns in residual correlations were observed, suggesting that these correlations were not specific to the set of predictor variables used to fit the model (Supplementary Results 2.1; Supplementary Figures S6).

4.2 Lagged climate variables did not improve prediction accuracy

The lagged climate model did not improve prediction accuracy for any taxon, which is unsurprising given the climatic stability of the late Holocene. The correlation between empirically estimated and predicted relative abundances remained approximately the same between the models including and not including climate conditions from the previous century (Table 1). Across all taxa, the median change in prediction bias between models (from the model with contemporaneous climate to the lagged climate model) ranged from -0.08 to 0.04 fraction of total stems, with most taxa experiencing changes in prediction accuracy of $\approx \pm 0.02$. There were no discernible patterns in change in prediction accuracy over space or time, except that bias changed (either worsening or improving) most where the taxon was most relatively abundant.

4.3 High climate sensitivity at biome boundaries

In general, community composition exhibited low sensitivity to temporal change in climate conditions (i.e., near-zero slopes between climate and relative abundance in most grid cells; Figure 4; Supplementary Figures S23 & S24). However, despite the relative stability of climate and vegetation, taxon relative abundances were on average more sensitive to climate change near the region's ecotones, at the edges of the savanna biome. Relative to a background sensitivity of approximately zero, oak climate sensitivity was higher in the grid cells at the northern and eastern boundaries of the savanna biome (range: -2.2-1.5; Figure 4). Higher precipitation seasonality favored increasing oak relative abundance (maximum sensitivity: 1.5) to the north at the expense of pine (minimum sensitivity: -0.97), the dominant taxon of the Minnesota northern pine forests (Figure 1b), and other hardwood taxa (minimum sensitivity: -0.93). Meanwhile, oak relative abundance decreased with higher precipitation seasonality to the south (minimum sensitivity: -2.2), with elm (the most relatively abundant tree taxon in the tallgrass prairie) having a corresponding, although smaller, increase in relative abundance with increasing precipitation seasonality (maximum sensitivity: 0.80). Similar, though smaller, spatial and taxonomic patterns in climate sensitivity existed near the boundaries of the savanna biome for average annual temperature (Supplementary Figure S23) and total annual precipitation (Supplementary Figure S24). Sensitivity to average annual temperature ranged from -0.11 (pine) to 0.071 (oak), while sensitivity to total annual precipitation ranged from -0.13 (pine) to 0.099 (oak), both along the northern savanna-forest boundary.

4.4 Autocorrelation explains increasing hemlock relative abundance

Using our autoregressive model, we found that across our entire geographic domain, the coefficient describing the relationship between previous and current hemlock relative abundance (median β_{abund} [95% CrI] = 0.23 [0.22,0.23]) was greater than the coefficients between hemlock relative abundance and any of the climate and soil covariates (β_{aat} = -0.090 [-0.10,-0.075]; β_{tpr} = 0.15 [0.14,0.17]; β_{prsd} = -0.078 [-0.088,-0.069]; β_{sand} = 0.013 [0.022,0.023]) after standardizing all covariates. It might be expected that any taxon's relative abundance should be strongly related to its previous relative abundance. However, fitting our autoregressive model only to north-central Wisconsin, where our JSDMs failed to predict trends in community composition, provides further evidence in support of a amplified vegetation response in this region. As expected, the covariates describing the relationships between environmental conditions and hemlock relative abundance were much weaker because there was less environmental variability in this smaller geographic region (β_{aat} = -0.050 [-0.12,0.013]; β_{tpr} = -0.017 [-0.061,0.026]; β_{prsd} = -0.041 [-0.12,0.038]; β_{sand} = -0.082 [-0.16,-0.0090]). On the other hand, the coefficient describing the relationship between previous and current hemlock relative abundance was much higher in north-central Wisconsin than our full geographic domain (β_{abund} = 0.39 [0.34,0.44] versus 0.23 [0.22,0.23] for the full domain), suggesting that small temporal changes in climate conditions in this region did not explain the observed increase in hemlock relative abundance.

5 Discussion

Empirically estimating the long-term sensitivity of vegetation to climate using paleoecological records is critical for assessing the assumptions in models predicting vegetation change from current climate trends (Anderson et al., 2006; Swetnam et al., 1999). Toward this aim, our investigation makes several important advances. We are among the first to use independent estimates of climate and tree community composition (i.e., not pollen composition) to quantify the climate-vegetation relationship over ecologically-relevant timescales for long-lived trees (Dawson et al., 2019; Paciorek and McLachlan, 2009; Prentice et al., 1991). We then use these reconstructions to evaluate a common assumption, that climate and vegetation were in equilibrium during the relatively stable late Holocene. We primarily found support for climate-vegetation equilibrium over the last 2000 years, but identified notable exceptions at biome and species range boundaries. Among those exceptions, we found no strong evidence for lagged vegetation responses, but rather found that these changes are consistent with amplified vegetation change.

5.1 Widespread support for climate-vegetation stability

The last 2000 years of the pre-Industrial Holocene in the UMW were characterized by relatively stable climate, and generally stable tree community composition. Aligning with previous work (Prentice et al., 1991; Williams et al., 2020), we found that the climatic stability allowed plant community composition to be in dynamic equilibrium with climate. This is important because it suggests that community composition was by and large a predictable function of climate conditions over centennial time scales (Prentice et al., 1991; Siefert et al., 2012). Furthermore, our success in predicting patterns of community composition with only a few, widely available climate drivers is encouraging for making predictions of vegetation change with available climate projections (Maloney et al., 2014). Our findings suggest that predictions predicated on the vegetation-climate equilibrium assumption will be particularly successful if the pace of climate change levels off, as is expected under lower greenhouse gas emissions scenarios (Calvin et al., 2023).

In our JSDM, which assumes that a change in climate leads to a corresponding change in community composition, climate and soil conditions were able to explain most variability in community composition. The insignificant residual correlations between most taxon pairs (Figure 3), in combination with strong relationships between climate covariates and taxon relative abundances (Figure 2a-c), indicate that climatic and edaphic variables were sufficient to explain most variability in community composition across space and time. Furthermore, according to our sensitivity analysis, we found that community composition was relatively insensitive to temporal changes in precipitation seasonality and even less sensitive to temporal changes in average annual temperature and total annual precipitation over most of the UMW (Figure 4 and Supplementary Figures S23 & S24). Slopes describing the relationship between each climate variable and each taxon's relative abundance over time were near zero for most grid cells, highlighting that vegetation change was negligible as a consequence of limited directional climate change.

5.2 Limited evidence for lagged vegetation response

In regions that were not in equilibrium with climate, we did not find support for lagged vegetation change. Previous research has emphasized the role of lagged vegetation change in the early- to mid-Holocene (Davis et al., 1998; Woods and Davis, 1989), when the pace of climate change was faster than the late Holocene (Davis, 1984). However, the extent to which vegetation change continued to lag climate change in the late Holocene was less certain. When we included previous climate conditions as covariates in our JSDM, predictive performance stayed approximately the same (Table 1), suggesting that previous trends in climate did not hold additional explanatory power over climate of the current time step, although vegetation responses to previous, discrete climate events would not be captured using our simulated climate estimates. Our findings do not suggest that vegetation *never* lags climate when climate is changing more rapidly (Williams et al., 2020; Davis, 1984). In fact, it is expected that lagged vegetation responses will be more prevalent in the future under anthropogenic climate change, for instance, than we identified in the late Holocene (Sharma et al., 2022; Block et al., 2022).

5.3 Amplified community change at biome and species range boundaries

We found that climate gradients were insufficient for explaining community composition variability over space and time in two key instances: (1) at a biome boundary and (2) at a species range boundary. Over space, our climate and soil texture variables failed to explain distinct prairie and savanna and forest and savanna biomes that existed in close proximity and similar climate space at ecotones. Over time, shifts in community composition along the savanna ecotones and the rise in hemlock abundance in northern Wisconsin were not explained by directional change in climate over the last two millennia. We infer that these patterns in community composition are more consistent with amplified that equilibrium vegetation change. This has potentially important implications for predicting vegetation change in response to future climate change at biome and species range boundaries. Change in tree species abundances and distributions at extant savanna-forest ecotones and at species range boundaries may be better anticipated when accounting for processes that can give rise to vegetation-climate disequilibrium, such as vegetation-environment-disturbance feedbacks, in predictive models (Holt, 2009).

We recognize that our identification of amplified vegetation change is dependent on our use of simulated, downscaled climate covariates, which likely miss some finer scale climate trends (Kaufman and Broadman, 2023). However, the trends in amplified vegetation change that we detected are unlikely to be exclusively attributed to the limitations of our climate simulations. If these finer scale climate trends were responsible for vegetation change, then the spatial smoothness of Midwestern climate would be expected to produce more widespread change in vegetation, rather than localized at biome and species range limits. This expectation is supported by the gentle topography of the UMW region, which would limit localized climate phenomena (Daly et al., 2008). Nonetheless, future work should focus on combining climate model simulations with empirical climate reconstructions to overcome limitations to quantifying the climate-vegetation relationship stemming from uncertainty and bias in historical climate proxies and simulations (Tipton et al., 2016).

5.3.1 Savanna biome boundaries

The savanna-forest ecotone is one of the regions where Dawson et al. (2019) identified significant change in community composition over time. Our findings suggest that vegetation change was not solely driven by climate at the savanna biome's boundaries, where high residual correlations remained in our JSDM (Figure 3) and community composition was particularly sensitive to temporal changes in climate (Figure 4). A reasonable explanation for the non-equilibrium climate-vegetation relationship in this region is that vegetation-environment-disturbance feedbacks were drivers of community composition beyond the direct impact of climate (Williams et al., 2020; Heilman, 2019). At the savanna-forest boundary, savanna vegetation promotes– and is promoted by– higher frequency and intensity disturbance from wildfire, cultural fire, and megaherbivory than closed-canopy forest both in the historical UMW (Dey et al., 2017; Abrams and Nowacki, 2021; Abrams et al., 2021; Munoz et al., 2014; Tulowiecki and Larsen, 2015; Dey and Kabrick, 2015) and in the extant tropics (Staver et al., 2011; Flores et al., 2024; Bonan, 2016; Staver et al., 2021). Reciprocally, forest vegetation suppresses disturbance, allowing for the development of a denser canopy structure (another feedback) (Heilman, 2019; Nowacki and Abrams, 2008). Similarly, at the savanna-prairie boundary, the high prevalence of disturbances prevented the establishment of mature, disturbance-tolerant trees in the prairie, but once established in the adjacent savanna, oaks were able to persist despite frequent disturbance due to physiological adaptations to disturbance and cultivation by humans (Abrams, 1992; Abrams et al., 2021).

Despite our model's general ability to explain community composition by climatic gradients (Figure 2), the sharpness of the savanna ecotones was poorly explained by climate and soil conditions alone. This is supported by the negative residual correlations between oak and most other tree taxa in our JSDM (Figure 3), which indicate that our climatic and edaphic variables were insufficient for distinguishing tree community composition at the boundaries of the savanna biome (Clark et al., 2017; Roberts et al., 2022; Bystrova et al., 2021). The strongest negative correlations existed between oak and elm, and between oak and maple (Figure 3. Each of these taxa was the dominant tree taxon in its biome: oak was the dominant taxon in the savanna, elm in the prairie (Goring et al., 2016; Grimm, 1984), and maple in the southern hardwood forests directly east of the savanna (Figure 1b). These negative residual correlations indicate that under the same climate conditions, oak or other tree taxa could be abundant, but not both. While the residual correlations could be driven by a missing abiotic environmental covariate, we suggest that residual correlations arise from the existence of vegetation-environment-disturbance feedbacks is a parsimonious explanation a role of disturbance that is well documented (Bonan, 2016; Staver et al., 2011).

Our sensitivity analysis supports the conclusion that both climate and vegetation-environment-disturbance feedbacks drove vegetation change at UMW ecotones. We found that community composition was sensitive to small changes in all three of our climate variables near the boundaries of the savanna biome (Figure 4, Supplementary Figures S23 & S24). The higher sensitivity indicates that there were larger changes in community composition at the boundaries of the savanna biome than in adjacent regions within the forest and savanna biomes, despite relatively limited directional climate change. The heightened climate sensitivity was particularly pronounced for oak, and taxa most relatively abundant in the prairie (elm) and forest (pine and other hardwood taxonomic group) biomes directly adjacent to the savanna (Figure 4). This supports the assertion that vegetation-environment-disturbance feedbacks may lead to disproportionate community changes in response to relatively minor climate change along biome boundaries (Williams et al., 2020; Flores et al., 2024).

5.3.2 Hemlock species range limit

The rise in the relative abundance of eastern hemlock in forest communities at its western range limit represents one of the largest and most surprising changes in community composition in the UMW over the last 2000 years (Dawson et al., 2019; Parshall, 2002). We found no evidence that this trend was due to directional climate change, and, instead, propose that the increase in hemlock relative abundance was driven by a combination of climate and vegetation-environment feedbacks. Specifically, the demographic momentum exhibited by hemlock is consistent with neighborhood effects, processes that promote self-replacement and proliferation of a tree species at the expense of other taxa that share a similar climatic niche (Milenkowic, 2024; Frelich and Reich, 1999). As a late-successional dominant, neighborhood effects could have included increased hemlock representation in the seed bank, shading of less shade tolerant competitors, mesophication of the understory, and alteration of soil chemistry (Davis et al., 1992; Frelich and Reich, 1999; Davis et al., 1998; Holt, 2009; Braziunas et al., 2025). Previous research has demonstrated that neighborhood effects promote the development and maintenance of distinct hemlock-dominated and hardwood-dominated forest patches in the UMW (Davis et al., 1992; Frelich and Reich, 1999; Milenkowic, 2024). Our results additionally suggest that vegetation-environment feedbacks can add up to affect community composition at the landscape scale, such as in northern Wisconsin.

Using our JSDM, we found weak negative residual correlations between hemlock and pine, hemlock and spruce, and hemlock and tamarack (Figure 3). This highlights that, similar to taxa at the savanna ecotones, environmental conditions alone were unable to explain the abundances of forest tree taxa where hemlock was relatively abundant at its range limit (Clark et al., 2017; Poggiato et al., 2021; Roberts et al., 2022). Rather, in similar climatic and edaphic space, hemlock or another conifer taxon (pine, spruce, or tamarack) could proliferate, but not together. Unlike the savanna-forest ecotone, however, our JSDM was unable to predict temporal dynamics in hemlock relative abundance from environmental covariates and residual correlations. Hemlock relative abundance was predicted with very high accuracy (bias ≈ 0) everywhere except in north-central Wisconsin, with the magnitude of bias increasing over time only in this region (Supplementary Figure S12). Our autoregressive model suggests that demographic momentum may be responsible for the community change in the northern Wisconsin region (Figure 1f). We found that current hemlock relative abundance was more sensitive to population lags than to spatiotemporal changes in climatic and edaphic conditions across the UMW. Furthermore, this effect was stronger when the model was fit only to north-central Wisconsin. This supports previous work at shorter timescales indicating that mature hemlocks are less sensitive to, and more equipped to withstand, moderate climate change (Bigelow et al., 2019) and competition (Woods, 2000) in the absence of stand-clearing disturbance than other UMW forest tree taxa.

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6 Box 1. Definitions of vegetation responses

- **Dynamic equilibrium**: Change in plant abundances and distributions is proportional to and concomitant with climate change over the spatial and temporal scales being studied. Dynamic equilibrium occurs when vegetation dispersal and demography is able to track the pace of climate change.

- **Lagged**: Change in plant abundances and distributions is proportional to, but lags behind, climate change because of processes including slow tree demography and dispersal limitations.

- **Amplified**: Change in plant abundances and distributions is *not* proportional to climate change (e.g., more rapid than climate change). Amplified vegetation change can result from a combination of climate and vegetation-environment feedbacks. Feedbacks are processes through which the vegetation modifies its environment, thereby making the environment more suitable for its own proliferation.

Table 1: **Correlations between observed and predicted taxon relative abundance from models with and without lagged climate variables**

Taxon	Without climate lags	With climate lags
Beech	0.69	0.65
Birch	0.71	0.70
Elm	0.57	0.54
Hemlock	0.62	0.60
Maple	0.40	0.40
Oak	0.75	0.75
Other conifer taxa	0.71	0.70
Other hardwood taxa	0.65	0.64
Pine	0.49	0.46
Spruce	0.68	0.67
Tamarack	0.70	0.66

Note: Correlations refer to Pearson product moment correlations between median observed and median predict taxon relative abundances, across 100 posterior draws of STEPPS and across 100 model predictions, respectively. Correlations were calculated irrespective of the spatial location of the grid cell and the time step of prediction.

7 Figure captions

Figure 1. Conceptual representation of the types of vegetation change (a), map of reconstructed vegetation in our study region (b), and conceptual representation of our hypothesis tests (c)-(f). (a) Given a climate forcing (black line), vegetation response can be in equilibrium with climate (blue), lag climate change (green), or be amplified (yellow). The dashed line separates the earlier Holocene (rapid change; left) from the late Holocene (stable climate; right). (b) The most relatively abundant taxon in each 24 km x 24 km grid cell at 300 years before present (YBP) across the Upper Midwest, USA (UMW). Encircled regions are locations where Dawson et al. (2016) found significant change in vegetation over the last two millennia. (c) Small residual correlations from our generalized joint attribute model (GJAM) indicate vegetation was in equilibrium with climate because current climate conditions explain patterns in community composition, while large correlations could indicate either lagged or amplified vegetation response. We investigate evidence of amplified and lagged vegetation responses in (d)-(f). (d) We fit our GJAM with current and previous climate conditions to test for lagged vegetation responses. Improvement to vegetation prediction accuracy after including previous climate would indicate a lagged vegetation response. (e) We tested whether there were patterns in the temporal climate sensitivity of taxa across space. Higher climate sensitivity is indicative of amplified vegetation change because the magnitude of vegetation change is greater than the near-0 vegetation change anticipated under no directional climate change (f) Amplified vegetation change could also be consistent with strong temporal autocorrelation. If community composition is more sensitive to previous community composition (*i.e.,* AR(1) term) than to climate conditions, this could be indicative of amplified vegetation change because of demographic momentum.

Figure 2. (a-c) Climate coefficient estimates from the GJAM. In each facet, the x-axis shows the

taxa in order from most negative to most positive median coefficient estimate. The y-axis shows the coefficient estimates standardized with respect to the independent variables and the latent response variable covariance. The violins show the distribution of coefficient estimates. Facets show coefficients for different covariates. Note the scale of the y-axis is different for each facet. Asterisks denote taxa that are primarily associated with open-canopy ecosystems. Note that soil texture (soil % sand) was also included as a covariate in the model and is not shown. (a) Average annual temperature, (b) total annual precipitation, (c) precipitation seasonality. (d) Joint sensitivity of all taxon relative abundances to each environmental covariate. Higher sensitivity indicates that the environmental covariate has a greater influence on community composition.

Figure 3. Correlation matrix showing residual correlations between taxon relative abundances, after accounting for the taxa's joint dependence on climate and soil texture. Color indicates median correlation between taxon pairs. Width of the ring depicts the 95% credible interval (CrI) in the correlation: the outer edge of the ring is the upper 97.5 quantile and the inner edge, the lower 2.5 quantile. Negative correlations are in red, positive correlations in blue. Correlations with 95% CrI overlapping zero were omitted from the figure for clarity.

Figure 4. The sensitivity of taxon relative abundances to precipitation seasonality. Sensitivity is defined as the standardized coefficient fitting the relationship between the precipitation seasonality and taxon relative abundance (facets) over time in a given grid cell. Values near zero indicate low sensitivity of relative abundance to precipitation seasonality. Larger absolute values (gold [positive] and blue [negative]) indicate that there was a greater change in relative abundance relative to the rate of change in precipitation seasonality.

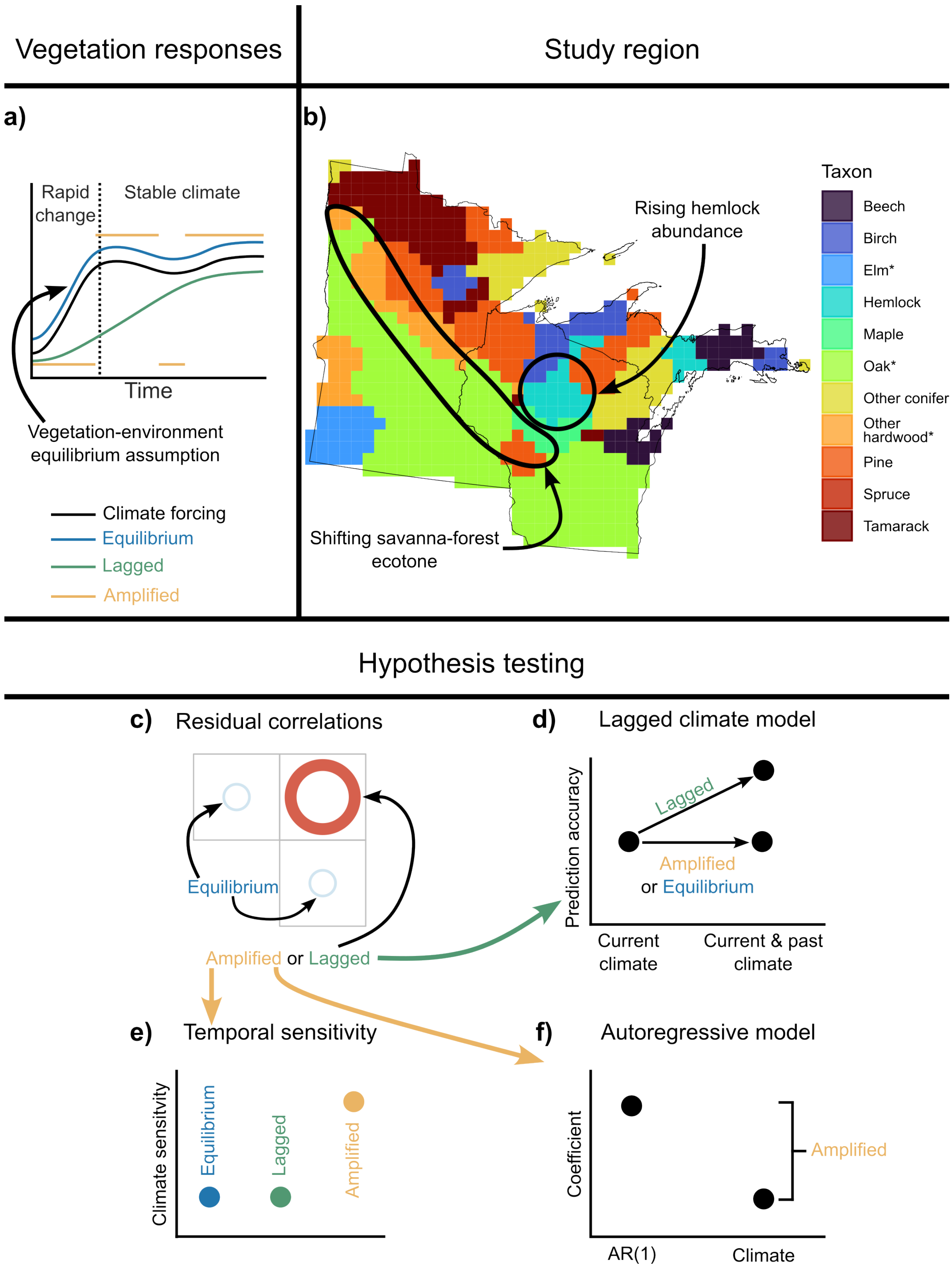
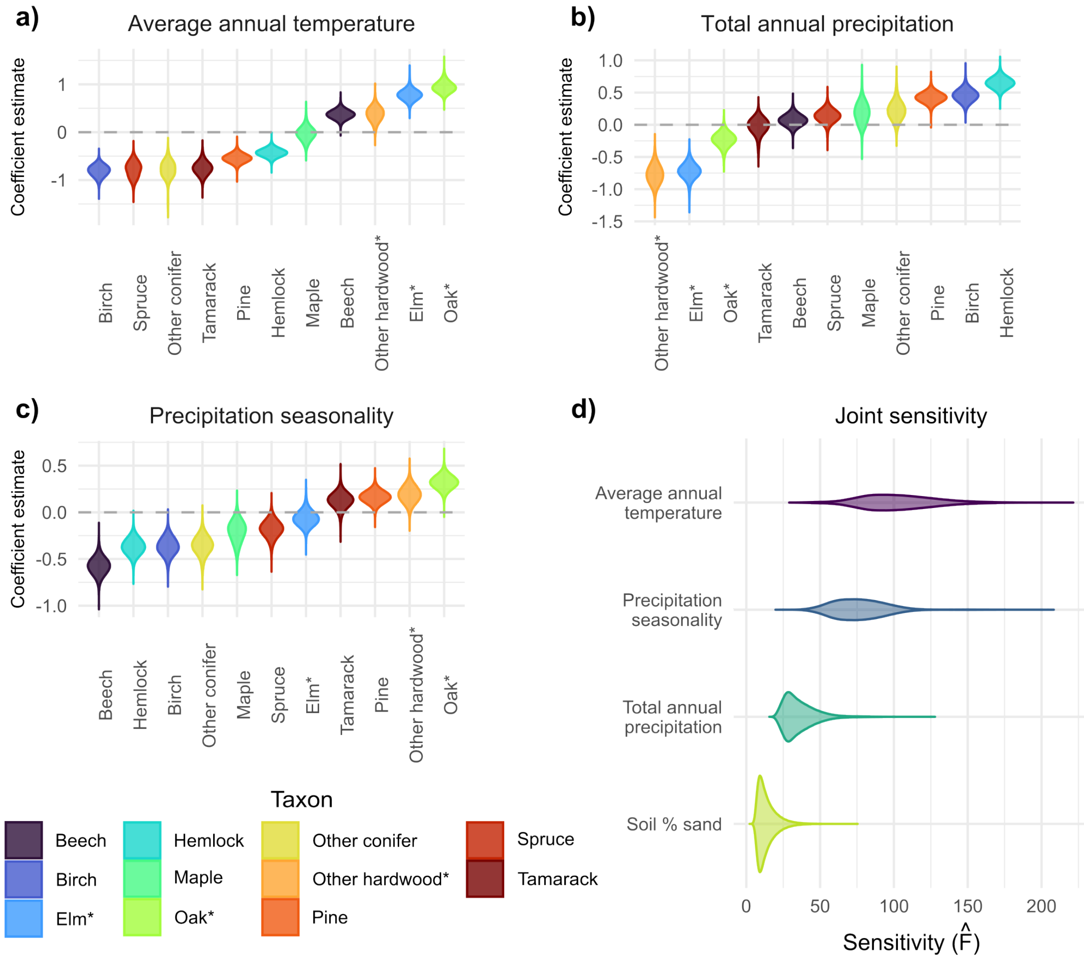

Vegetation responses
Study region
a)
Rapid change
Stable climate
Time
Vegetation-environment equilibrium assumption
Climate forcing
Equilibrium
Lagged
Amplified
b)
Rising hemlock abundance
Shifting savanna-forest ecotone
Taxon
Beech
Birch
Elm*
Hemlock
Maple
Oak*
Other conifer
Other hardwood*
Pine
Spruce
Tamarack
Hypothesis testing
c) Residual correlations
Equilibrium
Amplified or Lagged
d) Lagged climate model
Prediction accuracy
Lagged
Amplified
or Equilibrium
Current climate
Current & past climate
e) Temporal sensitivity
Climate sensitivity
Equilibrium
Lagged
Amplified
f) Autoregressive model
Coefficient
Amplified
AR(1)
Climate

Figure 1:



a)
Average annual temperature
Coefficient estimate
Birch
Spruce
Other conifer
Tamarack
Pine
Hemlock
Maple
Beech
Other hardwood*
Elm*
Oak*
b)
Total annual precipitation
Coefficient estimate
Other hardwood*
Elm*
Oak*
Tamarack
Beech
Spruce
Maple
Other conifer
Pine
Birch
Hemlock
c)
Precipitation seasonality
Coefficient estimate
Beech
Hemlock
Birch
Other conifer
Maple
Spruce
Elm*
Tamarack
Pine
Other hardwood*
Oak*
d)
Joint sensitivity
Average annual temperature
Precipitation seasonality
Total annual precipitation
Soil % sand
Sensitivity ($\hat{F}$)
Taxon
Beech
Birch
Elm*
Hemlock
Maple
Oak*
Other conifer
Other hardwood*
Pine
Spruce
Tamarack

Figure 2:


Birch
Elm
Hemlock
Maple
Oak
Other conifer
Other hardwood
Pine
Spruce
Tamarack
Beech
Birch
Elm
Hemlock
Maple
Oak
Other conifer
Other hardwood
Pine
Spruce
1
0.78
0.56
0.33
0.11
-0.11
-0.33
-0.56
-0.78
-1

Figure 3:

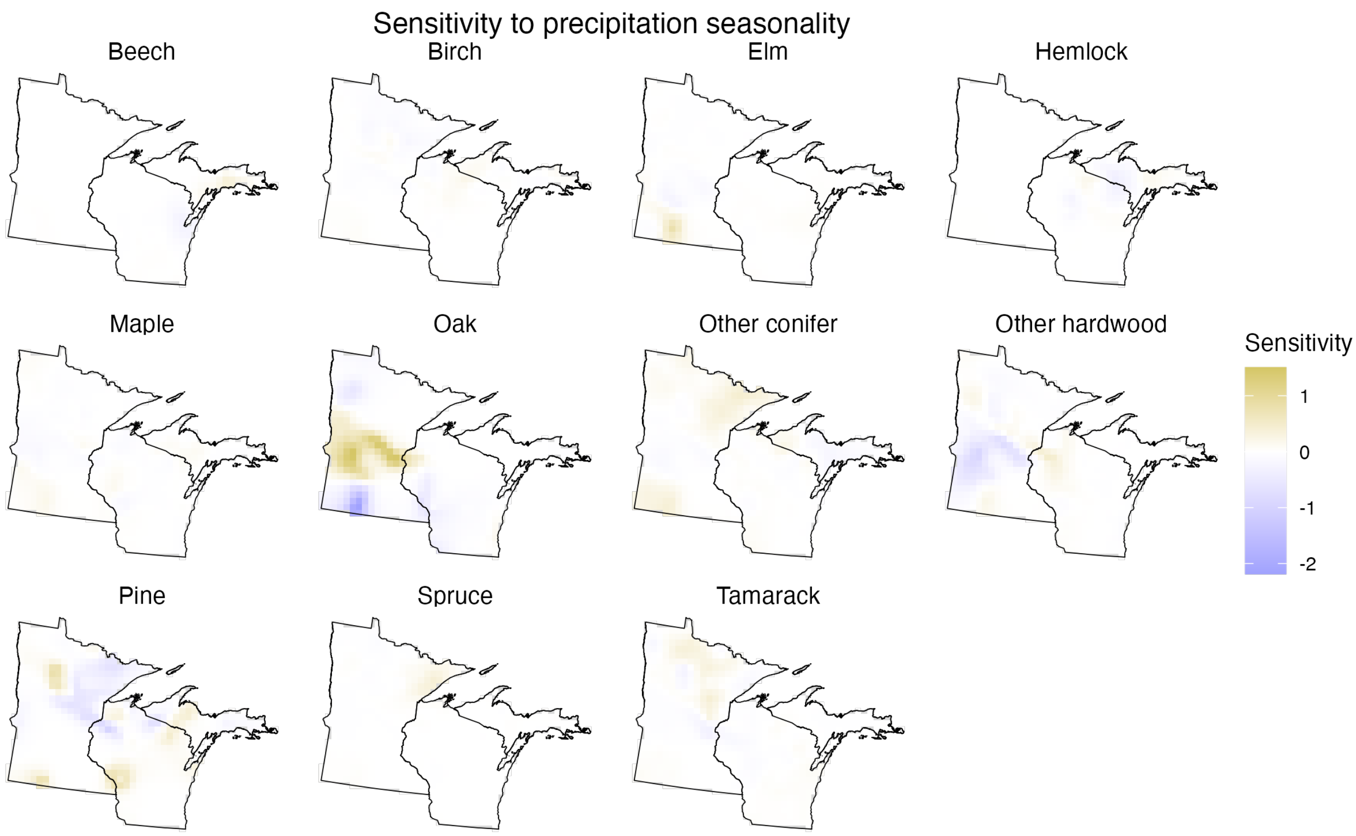

Sensitivity to precipitation seasonality
Beech
Birch
Elm
Hemlock
Maple
Oak
Other conifer
Other hardwood
Pine
Spruce
Tamarack
Sensitivity
1
0
-1
-2

Figure 4: